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Shape Optimization of Waveguides for Improving the Directivity of Soft Dome Tweeters

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ABSTRACT

This study presents a waveguide design strategy that integrates the modal matching method and the finite element method to align the spectral distributions of both on-axis and off-axis responses in a soft dome tweeter. Due to its high computational efficiency, the modal matching method is initially employed to derive a preliminary optimized waveguide. This initial design is then used as the starting geometry for further refinement via the finite element method, which offers greater flexibility in shape modification. Experimental results show that waveguides optimized using either method improve the tweeter's directivity. Furthermore, a strong correlation is observed between the optimization outcomes of the two methods.

1 Introduction

For direct-radiating loudspeakers, directivity becomes increasingly narrower with rising frequency, causing significant attenuation of the off-axis response at higher frequencies. This effect, commonly known as the beaming phenomenon [1-3], results in a reduced sweet spot, which may be a drawback in certain listening environments.

Acoustic waveguides, or horns, have proven to be effective for controlling directivity in loudspeakers, offering relatively constant beamwidth across a wide frequency range [4]. By optimizing their shape, waveguides can be designed to achieve specific directivity control characteristics. Most existing studies on waveguide shape optimization assume a plane wave source [5-7], typically focusing on scenarios where the waveguide is coupled with a compression driver.

In this article, we present a waveguide design approach that combines the modal matching method (MMM) [8] and the finite element method (FEM) to improve the directivity of a soft dome tweeter (described in Section 2). We first define a specific optimization objective to address the directivity issue for the soft dome tweeter. Next, we perform preliminary shape optimization using the MMM, which effectively improves off-axis attenuation. Finally, the preliminary waveguide is further refined using FEM-based optimization, and the optimized design meets the target objectives. Overall, the optimization process demonstrates a progressive improvement in the tweeter's directivity.

2 Design Objective

Figure 1 shows the soft dome tweeter used in this study. Figure 2 presents its responses normalized to the on-axis at off-axis angles of 10°, 20°, and 30°, measured at a distance of 1 meter in a semi-anechoic chamber. The beam narrowing significantly increases off-axis attenuation, meaning that even a small deviation from the on-axis direction results in different frequency components being heard.

Figure 1. The soft dome tweeter used as the sound source in this study.

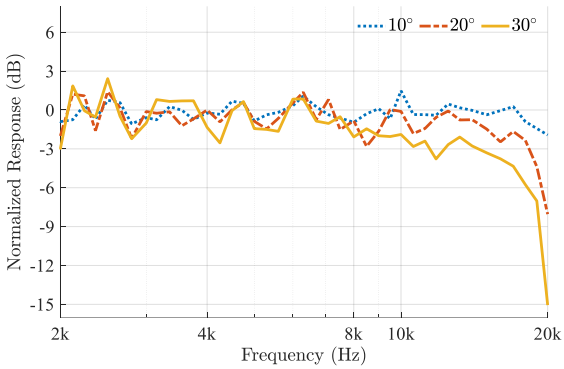


Figure 2. Measured responses normalized to the on-axis at off-axis angles of 10°, 20°, and 30° for the soft dome tweeter.

To reduce the frequency response difference within the horizontal 30° off-axis range, the design objective is given by

$$\begin{aligned} &\left|L_p(0^\circ, f) - L_p(\theta, f)\right| \leq 3 \text{ dB}, \\ &|\theta| \leq 30^\circ, 2 \text{ kHz} \leq f \leq 20 \text{ kHz}. \end{aligned} \tag{1}$$

Here, $L_p(\theta, f)$ represents the sound pressure level at a horizontal off-axis angle θ and frequency f, with all sound pressure levels measured at a distance of 1 meter. With the design objective outlined in Eq. (1), the optimized waveguide will extend the sweet spot for the tweeter.

3 Preliminary optimization Based on Mode Matching Method

The mode matching method (MMM), also known as the discrete multimodal method [9], has proven to be an efficient approach for solving sound propagation in waveguides, offering both high accuracy and computational efficiency. Building on the MMM, Dong et al. [10] proposed a method for optimizing the far-field directivity of an infinitely baffled axisymmetric waveguide. This method models the waveguide as a discrete structure consisting of N uniform ducts, as shown in Fig. 3. In each uniform duct with radius r, the axial velocity u can be expressed as $u = \mathbf{U}^{\mathrm{T}}\mathbf{\Psi}/S$, where $S = \pi r^2$ is the cross-sectional area, $\mathbf{\Psi}$ is a vector of the transverse modes, and $\mathbf{U}$ is a vector of the corresponding modal amplitudes. Through the discrete model, the relation between the velocity vector $\mathbf{U}_{\mathrm{m}}$ at the waveguide mouth and $\mathbf{U}_{\mathrm{t}}$ at the waveguide throat can be derived as

$$\mathbf{U}_{\mathrm{m}} = \left(\prod_{i=1}^{N} \mathbf{K}_i\right)\mathbf{U}_{\mathrm{t}}, \tag{2}$$

where $\mathbf{K}_i$ is a matrix related to the radii of the ducts from the ith to the Nth. If the waveguide throat is fed with a planar piston source, that is, $\mathbf{U}_{\mathrm{t}} = [1,0,0,\cdots]^{\mathrm{T}}$, then by defining $\mathbf{A}_l = \mathbf{K}_N\mathbf{K}_{N-1}\cdots\mathbf{K}_{N-l+1}$ and $\mathbf{B}_l = \mathbf{K}_l\mathbf{K}_{l-1}\cdots\mathbf{K}_1$ for $1 \leq l \leq N$, the gradient of $\mathbf{U}_{\mathrm{m}}$ with respect to the design variables r_j $(1 < j < N)$ is given by [10]

$$\frac{\partial \mathbf{U}_{\mathrm{m}}}{\partial r_j} = \left[\sum_{l=2}^{j}\left(\mathbf{A}_{N-l}\frac{\partial \mathbf{K}_l}{\partial r_j}\mathbf{B}_{l-1}\right) + \mathbf{A}_{N-1}\frac{\partial \mathbf{K}_1}{\partial r_j}\right]\mathbf{U}_{\mathrm{t}}, \tag{3}$$

and the design variables can be represented by a vector $\mathbf{r} = [r_2, r_3, \cdots, r_{N-1}]^{\mathrm{T}}$.

For an infinitely flanged waveguide mouth with the radius $r_{\mathrm{m}} \ll 1$ m, the directivity function is given by [10]

$$D(\theta) = \mathbf{U}_{\mathrm{m}}^{\mathrm{T}}\mathbf{\Upsilon}, \tag{4}$$

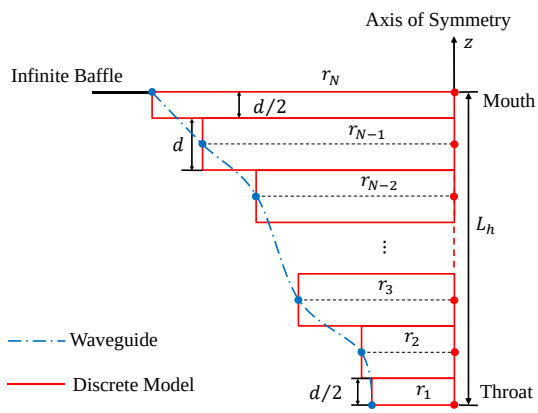


Figure 3. Discrete model of an axisymmetric waveguide mounted on an infinite baffle.

where $\boldsymbol{\Upsilon}$ is a vector of

$$\Upsilon_n(\theta) = \frac{\mathrm{j}k^2\rho_0 c r_\mathrm{m} \sin\theta\, J_1(kr_\mathrm{m}\sin\theta)}{\pi[(kr_\mathrm{m}\sin\theta)^2 - \gamma_n^2]}. \quad (5)$$

In Eq. (5), ρ_0 and c represent the density and sound speed of air, respectively, while k is the wavenumber, and γ_n denotes the nth zero of J_1, the first-order Bessel function of the first kind.

The objective function for directivity control is defined as [10]

$$G_1(\mathbf{r}) = \frac{1}{N_\theta N_f}\sum_{j=1}^{N_\theta}\sum_{i=1}^{N_f}\left[\hat{L}(\theta_j; f_i) - \frac{1}{N_f}\sum_{l=1}^{N_f}\hat{L}(\theta_j; f_l)\right]^2, \quad (6)$$

where $\hat{L}(\theta)$ is the sound pressure level normalized to the on-axis at off-axis angle of θ, given by

$$\hat{L}(\theta) = 10\log_{10}|D(\theta)/D(0)|^2. \quad (7)$$

Minimization of the objective function G_1 enables the maintenance of a constant attenuation at the off-axis angles θ_j ($j = 1,2,\cdots,N_\theta$) within the frequency range described by sampled frequencies f_i ($i = 1,2,\cdots,N_f$). Nevertheless, this accomplishment only addresses part of the design objective as depicted in Eq. (1).

To meet the requirement of attenuation less than 3 dB, we have additionally incorporated another objective function, which is given by

$$G_2(\mathbf{r}) = \frac{1}{N_f}\sum_{i=1}^{N_f}\left[\hat{L}(\theta_0; f_i) + \Delta L_p\right]^2, \quad (8)$$

where ΔL_p is an attenuation value less than 3 dB, and θ_0 is equal to 30° according to Eq. (1). G_2 ensures that the attenuation in the 30° off-axis direction is below 3 dB, while G_1 maintains constant off-axis attenuation for $\theta < 30^\circ$. The combination of these two functions results in the overall objective function

$$G(\mathbf{r}) = G_1(\mathbf{r}) + \mu G_2(\mathbf{r}), \quad (9)$$

where μ is an adjustable weighting coefficient. Compared with G_1, the new objective function G aligns more closely with the design objective outlined in Eq. (1). Thus, the design problem is a non-linear least squares optimization problem and can be solved with the Levenberg-Marquardt algorithm.

4 Further Optimization Based on Finite Element Method

Although the MMM offers high efficiency, the requirement for cross-sectional invariance limits the degrees of freedom in shape optimization. Therefore, COMSOL Multiphysics is introduced for further optimization, with the waveguide optimized using the MMM serving as the initial structure. Considering the symmetry, a quarter-model was created, including the tweeter and waveguide, as shown in Fig. 4. This full 3D electro-vibroacoustic multi-physical model consists of acoustic, structural, electromagnetic modules. The waveguide, mounted on an infinite baffle, radiates into a hemispherical air volume, with a perfectly matched layer (PML) surrounding the outside.

The further optimization through FEM mainly depends on the shape optimization module in COMSOL. The optimization variables $\mathbf{d} = \left[d_1, d_2, \cdots, d_{3N_\mathrm{wg}}\right]^\mathrm{T}$ are the displacements of the mesh nodes on the waveguide wall surface in the X, Y, and Z directions. N_wg is the number of the mesh nodes. The objective function at the frequency f_i is defined as

$$\mathrm{obj}(f_i,\mathbf{d}) = \left[\hat{L}(\theta_0; f_i) + \Delta L_p\right]^2, \quad (10)$$

where $\hat{L}(\theta_0; f_i)$ is the sound pressure level normalized to the on-axis at off-axis angle of 30° in the horizontal direction. The overall optimization objective is given by

$$\min_{\mathbf{d}}\left\{\max_i[\mathrm{obj}(f_i,\mathbf{d})]\right\}, \quad i = 1,2,\cdots,N_f \quad \text{s.t. } 0 \le \|\mathbf{d}\|_\infty \le d_l, \quad (11)$$

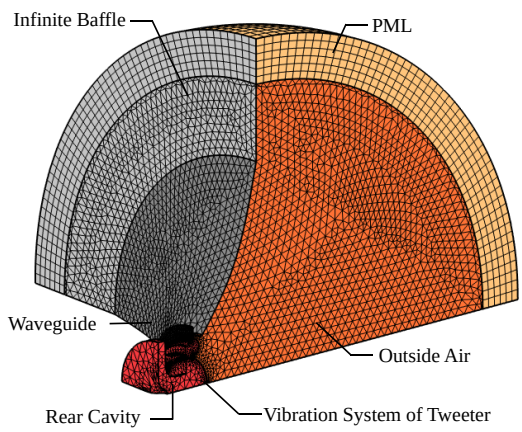


Figure 4. FEM model of the waveguide and tweeter.

where d_l is the maximum displacement of the mesh nodes.

5 Optimization Results

5.1 Simulation

Throughout the optimization process, a conical horn was selected as the initial waveguide, with radii of

$$r(z) = r_\mathrm{t} + \frac{(r_\mathrm{m} - r_\mathrm{t})z}{L_\mathrm{h}}, \tag{12}$$

where $r_\mathrm{t} = 16.14\ \mathrm{mm}$, $r_\mathrm{m} = 70\ \mathrm{mm}$ are the dimensions of throat and mouth, and $L_\mathrm{h} = 22.5\ \mathrm{mm}$ is the length of the waveguide.

In the preliminary optimization based on MMM, the excitation at the waveguide throat is approximately set as a planar piston source. The off-axis angles in the objective function G_1 are 10°, 20°, and 30°. The attenuation value ΔL_p is 1 dB, and f_i are the frequencies sampled at $1/12$-octave intervals within the range of 2 kHz to 20 kHz. The weighting coefficient μ is set to 0.02. The iteration stops when the relative change in the objective function value or the variable is less than 1×10^{-6}. In the further optimization based on FEM, the maximum displacement d_l is set to $6\ \mathrm{mm}$. Considering time cost, the optimization stops after 10 iterations.

Fig. 5 shows the waveguide geometry optimized based on the MMM. The contraction part below the throat is employed to ensure that the area of the waveguide throat is consistent with the radiation area of the tweeter. The waveguide shape obtained through further optimization based on FEM is shown in Fig. 6. The contour lines represent the deformation (in mm) of the further optimized shape relative to the shape optimized by MMM.

Fig. 7 shows the simulated responses at a horizontal off-axis angle of 30° for the tweeter with or without waveguides. For the initial waveguide, the attenuation of the tweeter in the 30° off-axis direction is improved at high frequencies, but still fails to meet the design objective. For both optimized waveguides, the attenuations relative to the on-axis response in the 30° off-axis direction are controlled below 3 dB within the frequency range from 2 kHz to 20 kHz. Additionally, compared with the waveguide optimized by MMM, the normalized response of the waveguide optimized by FEM shows better constancy, particularly within the high-frequency range above 10 kHz.

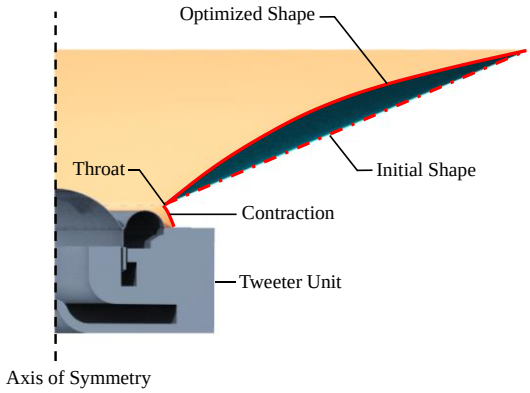


Figure 5. Initial shape and preliminary optimized shape of the waveguide connected with the tweeter unit.

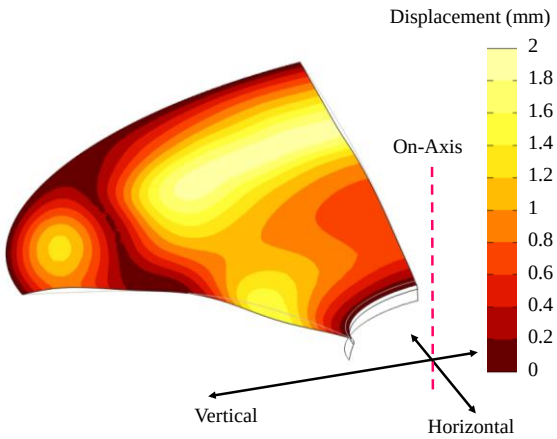


Figure 6. Further optimized waveguide geometry (1/4 model) with contour lines indicating the displacement (in mm) of the deformation relative to the shape optimized by MMM.

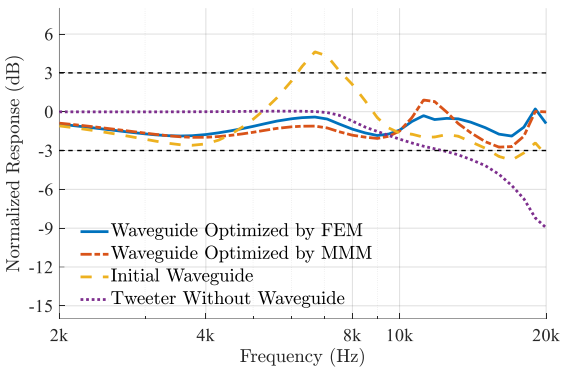


Figure 7. Comparison of simulated responses, normalized to the on-axis response, at a horizontal off-axis angle of 30° for the tweeter with or without waveguides.

5.2 Experiment

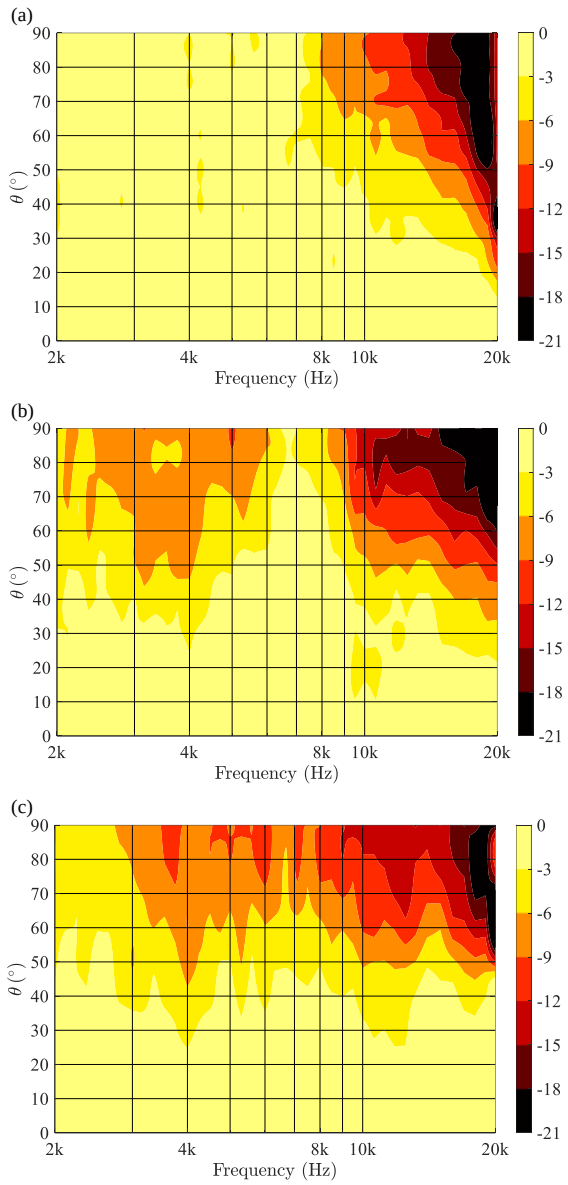
To check the performance of the optimized waveguides, both the waveguide optimized by MMM and the waveguide optimized by FEM were fabricated via 3D printing, and tested in a semi-anechoic chamber. The test signal input to the tweeter unit is a stepped sine signal generated by an audio interface (Fireface 400, RME, Germany), and amplified by a power amplifier (IP 450, Lab Gruppen, Germany). The directivity patterns of the tweeter with or without waveguides are presented in Fig. 8. Compared with Fig. 8(a), it can be seen from Fig. 8(b)(c) that the beaming effect has been mitigated after coupling the tweeter with the waveguides optimized using either the MMM or the FEM. Furthermore, the directivity of the waveguide optimized by FEM exhibits better constancy when compared with the waveguide optimized by MMM, especially at high frequencies above 10 kHz. This experimental finding shows good agreement with the simulated outcomes presented in Fig. 7.

Fig. 9 shows the measured responses at a horizontal off-axis angle of 30° for the tweeter with or without waveguides. In accordance with the simulation result, the measured curve of the initial waveguide exhibits an undesired fluctuation within the mid-frequency range. While the waveguide optimized by MMM effectively mitigates the midrange fluctuation, there are still some deficiencies in the measured response. Specifically, the disparity between the on-axis and the 30° off-axis responses at some frequencies (such as 4 kHz, 12 kHz and 20 kHz) has exceeded 3 dB. For the waveguide optimized by FEM, it can be seen that the measured 30° off-axis attenuation is less than 3 dB in the frequency range from 2 kHz to 20 kHz, which has met the design objective described by Eq. (1).

In addition to the effect of reducing off-axis attenuation, the optimized waveguide also improves the constancy of the attenuation curve with frequency variations. Table 1 lists the standard deviations of the curves in Fig. 9, which are calculated by

$$s = \sqrt{\frac{1}{N_f}\sum_{i=1}^{N_f}\left[\hat{L}(30^\circ; f_i) - \frac{1}{N_f}\sum_{j=1}^{N_f}\hat{L}(30^\circ; f_j)\right]^2}. \quad (13)$$

In Table 1, the standard deviation of the waveguide optimized by MMM is smaller than that of the tweeter without waveguide or with the initial waveguide, indicating that the optimized waveguide effectively improves the constancy of the 30° off-axis attenuation of the tweeter. Moreover, the standard deviation of the waveguide optimized by FEM is even smaller than that of the waveguide optimized by MMM. It implies that the constancy is further improved after the optimization based on FEM. This gradual enhancement in the measurement outcomes aligns well with the simulated results.



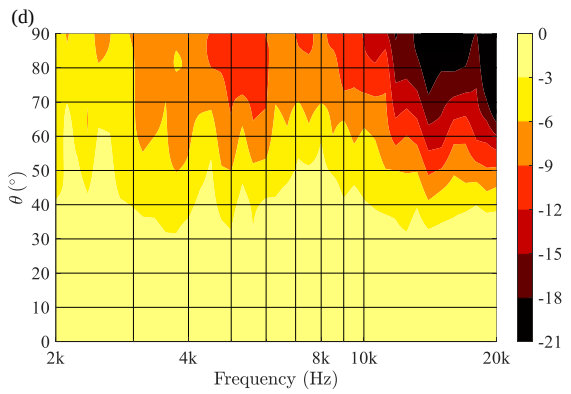


Figure 8. Directivity patterns of the tweeter (a) without waveguide, (b) with the initial waveguide, (c) with the waveguide optimized by MMM, and (d) with the waveguide further optimized by FEM.

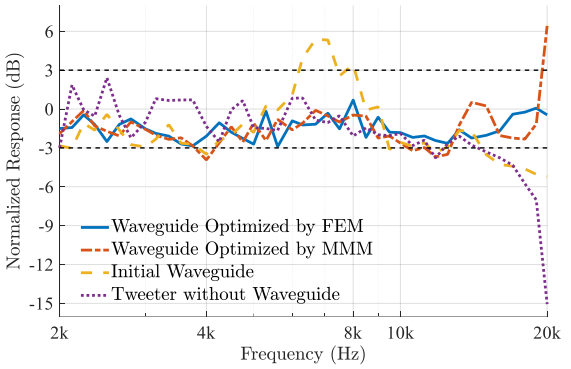


Figure 9. Comparison of measured responses, normalized to the on-axis response, at a horizontal off-axis angle of 30° for the tweeter with or without waveguides.

Table 1. Standard deviations of the 30° off-axis responses normalized to the on-axis.

	s
Tweeter without waveguide	2.48
Initial Waveguide	2.12
Waveguide optimized by MMM	1.50
Waveguide optimized by FEM	0.93

6 Conclusions

To improve off-axis response attenuation in the soft dome tweeter, incorporating a waveguide proves to be a simple yet effective solution. In this study, both MMM-based and FEM-based optimizations are applied to the waveguide shape design. As a result, across the 2 kHz to 20 kHz frequency range, off-axis attenuation within a ±30° range is kept below 3 dB. The optimized waveguide helps mitigate the beaming effect within this off-axis range, ensuring that the spectral distributions perceived at the listening positions remain essentially consistent.

For the axisymmetric waveguide model, the MMM transforms a two-dimensional (2D) problem into one-dimensional (1D) propagation of a set of transverse modes, effectively capturing the multimodal structure of guided-wave propagation and ensuring computational efficiency. The measured results show that the waveguide optimized using the MMM significantly improves the off-axis attenuation of the tweeter. Therefore, the MMM is well-suited for the preliminary shape optimization of waveguides, with the optimized shape serving as a starting point for FEM-based optimization. This approach requires fewer iterations compared to directly using a conical waveguide or similar structures as the initial shape, thus reducing the overall optimization time. Since the FEM-based optimization involves the displacement of each mesh node on the waveguide wall, it offers greater flexibility in shape modification compared to the MMM, leading to further improvements in the final design, which successfully meets the design objectives.

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